Course requirements

- Prerequisites
  - Data structures
  - Good working knowledge of Python programming
  - Linear algebra
  - Vector calculus

- Course does ***not*** assume prior imaging experience
  - computer vision, image processing, graphics, etc.

Course overview (tentative)

香港城市大學（東莞）
City University of Hong Kong (Dongguan)

We generally have 13 classes.

3 Classes
20/01/2026 - 03/02/2026

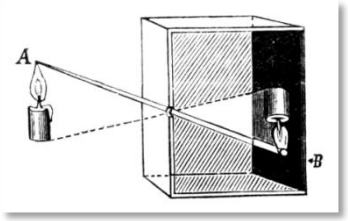
10 classes
03/03/2026 - 09/05/2026

	Empl ID	Name	Term	Class Nbr	Subject Area	Catalog Nbr	Description	Course Component	Week	Date	Weekday	Start Time
1	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	1	01/20/2026	Tues.	14:00
2	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	1	01/20/2026	Tues.	16:00
3	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	2	01/27/2026	Tues.	14:00
4	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	2	01/27/2026	Tues.	16:00
5	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	3	02/03/2026	Tues.	14:00
6	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	3	02/03/2026	Tues.	16:00
7	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	4	03/03/2026	Tues.	14:00
8	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	4	03/03/2026	Tues.	16:00
9	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	5	03/10/2026	Tues.	14:00
10	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	5	03/10/2026	Tues.	16:00
11	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	6	03/17/2026	Tues.	14:00
12	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	6	03/17/2026	Tues.	16:00
13	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	7	03/24/2026	Tues.	14:00
14	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	7	03/24/2026	Tues.	16:00
15	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	8	03/31/2026	Tues.	14:00
16	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	8	03/31/2026	Tues.	16:00
17	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	9	04/07/2026	Tues.	14:00
18	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	9	04/07/2026	Tues.	16:00
19	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	10	04/14/2026	Tues.	14:00
20	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	10	04/14/2026	Tues.	16:00
21	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	11	04/21/2026	Tues.	14:00
22	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	11	04/21/2026	Tues.	16:00
23	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	12	04/28/2026	Tues.	14:00
24	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	12	04/28/2026	Tues.	16:00
25	TBA005	TBA CS001	2520	1182	CS	5187	Vision and Image	Lecture	13	05/09/2026	Sat.	14:00
26	TBA005	TBA CS001	2520	1183	CS	5187	Vision and Image	Tutorial	13	05/09/2026	Sat.	16:00

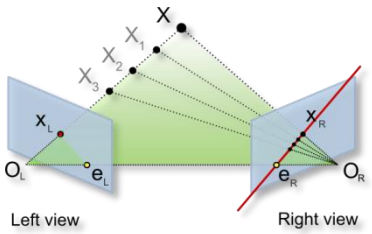
Course overview (tentative)

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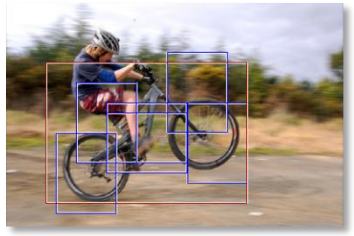
We generally have 13 classes.



1. Low-level vision (2-3 classes)
   - image processing, edge detection, feature detection, cameras, image formation



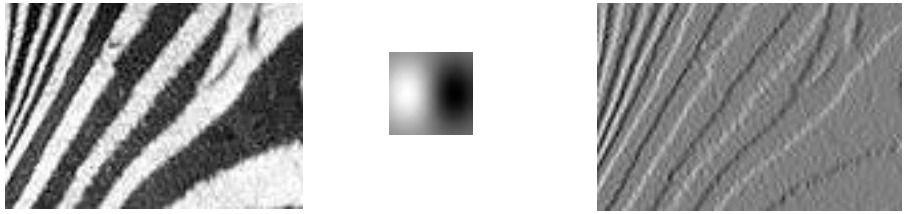
2. Geometry & appearance (3-4 classes)
   - projective geometry, stereo, structure from motion, optimization, lighting & materials



3. Recognition & generative models (6-8 classes)
   - object classification, deep learning, diffusion models

1. Low-level vision

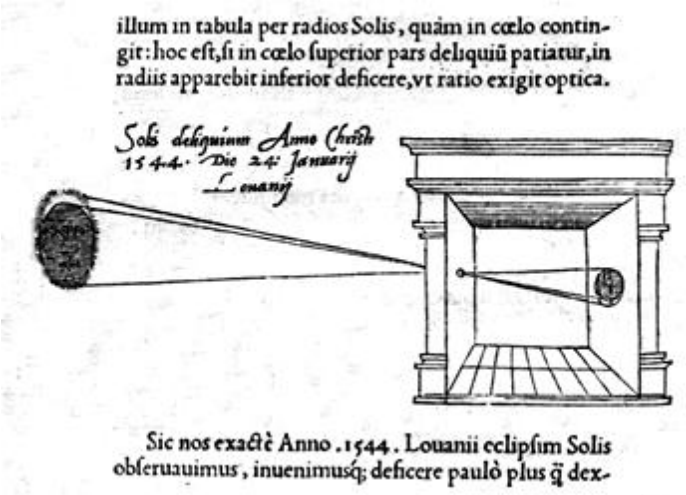
- Basic image processing and image formation



Filtering, edge detection



Feature extraction



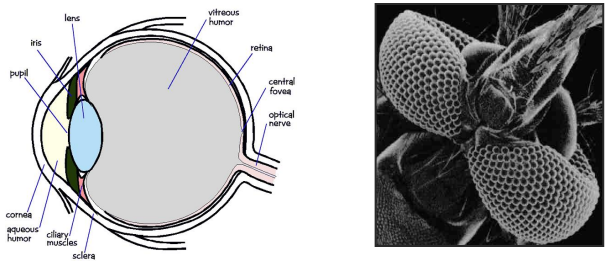
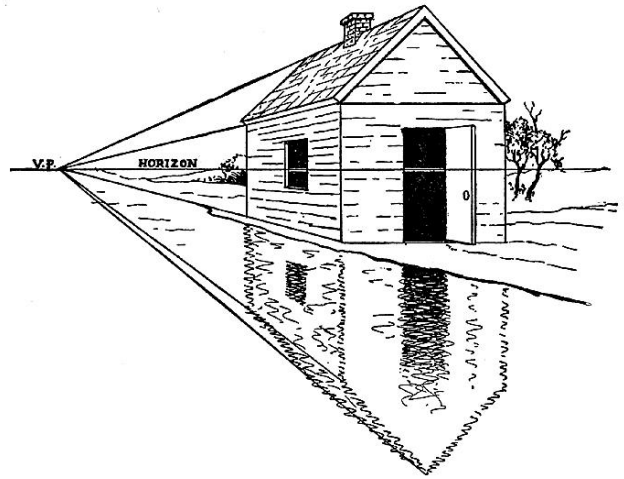


Image formation

2. Geometry & appearance



Projective geometry

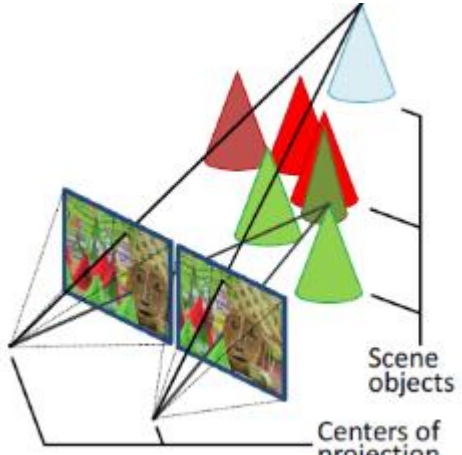
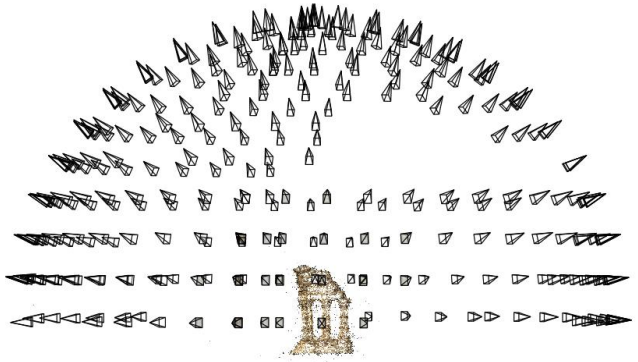


Image credit: IDS Imaging

Stereo vision



Multi-view stereo



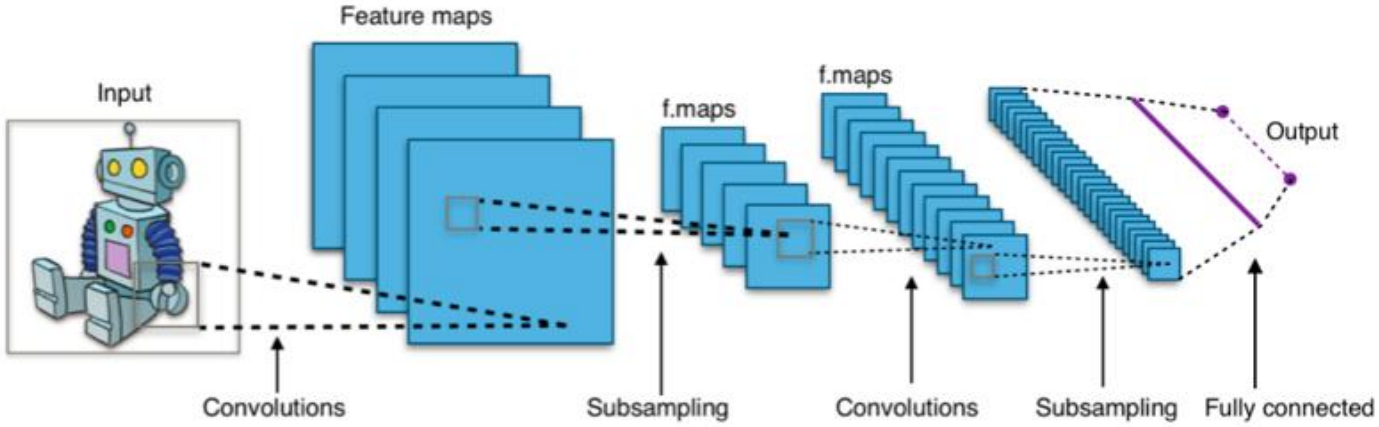
Structure from motion

3. Recognition & Generative Models



“dog”

Image classification



Convolutional Neural Networks



“a class watching a computer vision lecture”

Image generation

Lectures

- Lectures will be held in person in AC5-514
- If there is an instance where you cannot attend the class, please reach out to the instructor for approval
- No conversations during class, please!

Grading

- One midterm (between the 2nd and 3rd sections), and one final exam (in class)

- Grade breakdown (subject to minor tweaks):
  - Assignment : 40% = (Midterm: 30% + In-Class Quiz 10%)
  - Programming projects (2 projects): 10%
  - Final exam: 50%